EDSR-Based Super-Resolution of Multi-Look SAR Imagery

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***Abstract*—This paper presents a deep learning approach to super resolution of Synthetic Aperture Radar (SAR) images using the Enhanced Deep Super Resolution (EDSR) network. A single Capella space SLC SAR image of a desert scene is used for self supervised patch based training at a scale factor of 4x. The impact of multilook preprocessing on the training performance is investigated and shows that applying a 4 look average operation suppressed speckle noise that improves validation PSNR from 24.01 dB to 29.26 dB and SSIM from 0.2664 to 0.5667. Variance based patch filtering is applied to focus training of spatial informative patches. Results show the model can effectively reconstruct the high contrast terrain boundaries from the low resolution input and that the ground truth quality matters and preprocessing is a more significant factor than architectural design. Code for this project can be found on my github page[1]**

***Index Terms*—PyTroch, Deep learning, SAR, Multi-Look, EDSR, Super resolution**

I. Introduction

A. What is SAR?

Synthetic aperture radar (SAR) is a data collection method that involves instruments sending pulses of energy and recording the amount of energy that comes back after it has interacted with the surface [1]. SAR images are created with these reactions and changes in energy and physical structures like forests or mountains can be viewed, even moisture in soil [1]. In this project a SAR image of a desert is used and with the nature of SAR images being very large and in this case the shape of the image is 12197 x 38982 pixels.

B. Approach with very large images

With the SAR images so large an approach to handle this large amount of data. An image that is 12197 x 38982 is very large and a lot of memory and process power is needed process it. An approach on how to handle this is to split the image into low resolution patches [2] and have a deep learning AI model learn from them and upscale the images, and using a technique called multi look to increase both Peak Signal Noise Ratio (PSNR) and Structure Similarity Index Measure (SSIM) compared to simpler models. In figure 1 is an example of a lower resolution version of a patch and the ground truth of what the actual patch looks like.

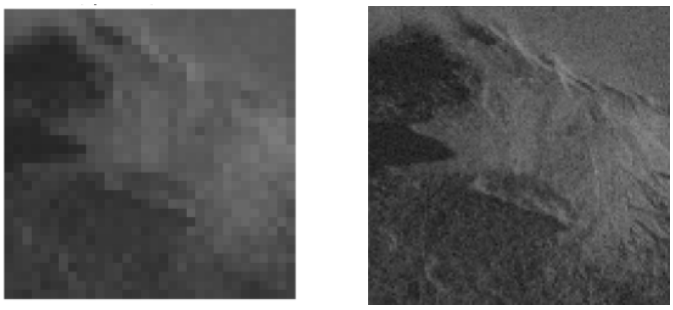

Fig. 1. Low-resolution (left) and high-resolution (right) SAR image patches

II. The deep learning and architecture

In order to reconstruct a high-resolution (HR) SAR image, the network architecture must be designed to try to preserve and exploit the spatial structure of the data , SAR images are characterized by the local intensity variations that corresponds to the physical surface property and the model has and must be able to capture the fine grained spatial dependencies while maintaining intensity information throughout the network.

One model architecture that addresses these requirements this through the use of 2D convolutional layers is ESDR. For SAR images this is not used since backscatter or grain in the images carries physical meaning and a bright pixel represents a strong reflecting surface and that information must be preserved throughout the network, so normalization of the data can't be done. One alternative to EDSR is Resnet which is a popular image generation architecture that has similar attributes but EDSR dose not use batch normalization whilst Resnet dose.

A. EDSR Architecture

The EDSR architecture uses 2D convolutional layers (conv2D) throughout the network and conv2D is well suited to these images because of the exploit of spatial locality. This implementation of EDSR follows a similar implementation done by Keras [3] but in this case is implemented in PyTorch instead of TensorFlow due to TensorFlow not supporting CUDA and windows any more.

- In figure 2 is a representation of the model where the first input layer is a Low resolution (LR) patch of the SAR

image. This image gets forwarded into the head conv2D layer.

- Head conv2D layer transforms raw pixel intensities into higher dimensional features that enables the network to represent local patterns such as edges and textures that are not explicitly visible in the original image.
- Res Block applies two conv2D layers in this implementation and with a layer of RELU activation and then adds the scaled result back into the input again. This is used for residual learning that allowed the network to focus on predicting refinements rather than a reconstructions of the entire signal. This improves training stability and enables deeper networks by mitigating the vanishing gradient problem. In this block the scaling factor is 0.1 that prevents each block form making larger updates and in term stabilizes training in deeper architectures.
- Later a pixel shuffle upsampling module that uses two sequential pixel shuffler to get depth to space operations rather than interpolation and each pixel shuffle takes a feature map with Cxr^2 channels and rearranges them into spatial map with C channels at rx the resolution. For the 4x scale used this is done in 2x steps rather than one 4x step. This is an advantage over bilinear or bicubic interpolations[x] is that upsampling kernel is learned form the data rather than fixed and allows the network to learn SAR upsampling patterns

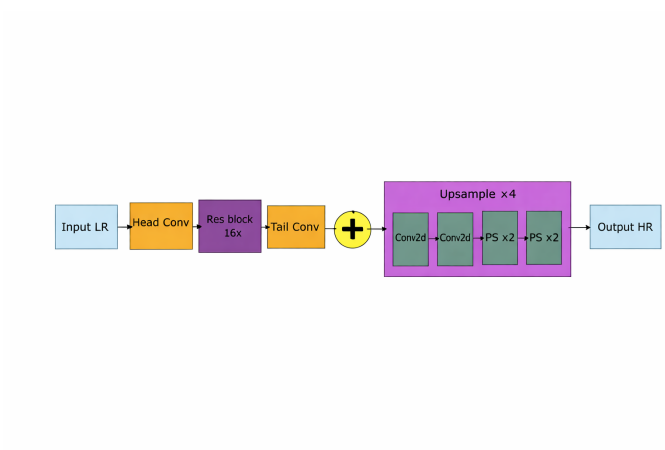

Fig. 2. EDSR Architecture

B. Loss Function

The model is trained using Mean absolute Error (MAE) loss, that i also referred to as L1 loss.

$$\mathcal{L}_{MAE} = \frac{1}{N}\sum_{i=1}^{N}|y_i - \hat{y}_i| \tag{1}$$

where y_i is the ground truth HR patch and $\hat{y}_i$ is the predicted output of the network, MAE was chosen over Mean Square error (MSE) for two reasons. MAE is less sensitive to outliers and with the high dynamic range of the SAR backscatter values even after compression, and in testing during this project has shown to create overly smooth images and it penalizes larger errors to hard.

III. Methodology

A. Dataset

The data set for this project consists of a single SAR image that is acquired by the Capella Space C13 satellites[4]. The image is a Single Look Complex (SLC) product with HH polarization, and has a shape of 12197 x 38982 pixels that covers a desert. SLC data represents the rawest from of SAR images, and can retain full resolution but a big problem is that it contains significant speckle noise due to the nature of radar images.

B. Preprocessing

1) MultiLook Processing: The raw data from SLC suffer from speckle noise as that appears as a salt and pepper pattern across the image as seen in figure 3. For super resolution training this speckle noise n the ground truth image comes with problems as it introduces unlearnable random components into the target signal and prevents the model from converging to meaning reconstructions.

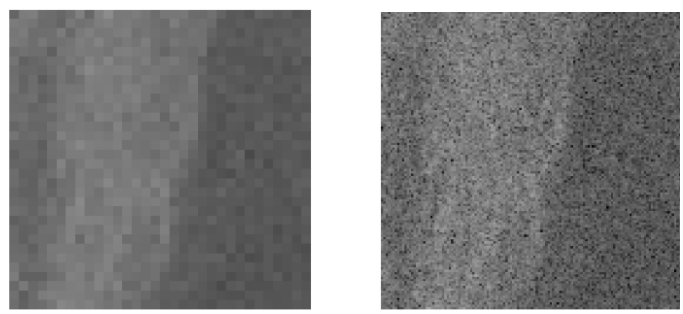

Fig. 3. Low-resolution (left) and high-resolution (right) SLC image patches

To address this an other method was used called 4-look multilook, and what it dose is that by averaging non-overlapping 4x4 pixel blocks across the image this caused the image to reduce the effective image dimensions to approximately 3049 x 9745 while significantly suppressing speckle noise. The result of this can be seen in figure 1. This in result produced a smoother ground truth patch that is more suitable for supervised training and the impact of this preprocessing step is demonstrated in the results where multilook processing improved PSNR from approximately 24 dB to 29 dB compared to training directly from the grainy SLC data.

2) Amplitude and Normalization: The input data is complex-valued SLC and the amplitude was extracted by computing the magnitude of each complex pixel. A logarithmic compression was applied using the log1p transformation

$$I_{compressed} = \log(1 + I_{amplitude}) \tag{2}$$

This compression reduced the dynamic range of the backscatter values which in this case can span many magnitudes and such making the data more suitable for neural

network training. The compressed image was normalized to the range [0 , 1] using min max normalization.

C. Patch Extraction

Since the SAR spatial dimension is so large training cant be performed in the whole image itself due to RAM constraints so training was performed on smaller patches extracted from the multilook image. HR patches of the size 128 x 128 pixels were extracted and using a sliding window with a stride of 64 pixels that produces overlapping patches that increase the effective number of training samples. Later a corresponding LR patch of size 32 x 32 pixels were generated by down sampling each HR patch by a factor of 4 using average pooling that uses a box-filter degradation model assumed by the network.

D. Variance filtering

With the scene being a desert a lot of the scene is homogeneous and regions with little spatial variations which provides limited useful training signal for super resolution. To fight this problem a bias towards more informative patches was applied and showed that only 40% of patches were retained for training. This reduced the total patches from 6946 to 2778 patches. This ensured the model was trained on meaningful patches containing spatial structures and such terrain with boundaries and geological features.

E. Train/Validation Split

The filtering patches were shuffled and split into a training set (85%) and a validation set (15%), yielding 2361 training patches and 417 validation patches. the validation set was used exclusively for monitoring generalization performance was not used for weights.

F. Data Augmentation

To increase some variety in the the limited patch count random horizontal and vertical flips were applied to training patches during loading. This increases the number of unique training samples by up to a factor of 4 without having to introduce physical unrealistic transformations.

G. Training Configuration

The model was trained for 100 epochs using the Adam optimiser [**?**] with an learning rate of 1×10^{-4}. The learning rate was reduced by of 0.5 when validation loss plateaued for 5 consecutive epochs using the ReduceLROnPlateau scheduler. Each epoch contained 147 gradient update steps with a batch size of 16, corresponding to one full pass through the training dataset. Training was performed on an NVIDIA GeForce RTX 4070 GPU with 12 GB of VRAM.

IV. Results

A. SLC VS Multilook

Table 1 presents the validation and SSIM score for the two configurations: training directly on the raw SLC data and training on the multilook preprocessed data and both configurations used the same EDSR architecture described before and same setup with the variance filter.

TABLE I
QUANTITATIVE COMPARISON OF SLC VS MULTILOOK TRAINING

Configuration	Val PSNR (dB)	Val SSIM
SLC (no preprocessing)	24.01	0.2664
Multilook (4 looks)	29.26	0.5667

When comparing both configurations there is a clear winner, the multilook achieved PSNR improved the PSNR score by 5.23 dB and SSIM score improvement of 0.3003 over the raw SLC baseline, this is a substantial gain alone from just changing preprocessing steps and no change in model architecture.

B. Qualitative comparison

Figure 4 and 5 show a visual comparison between Lr input, what EDSR has created and the HR ground truth patches for SLC and multilook configurations.

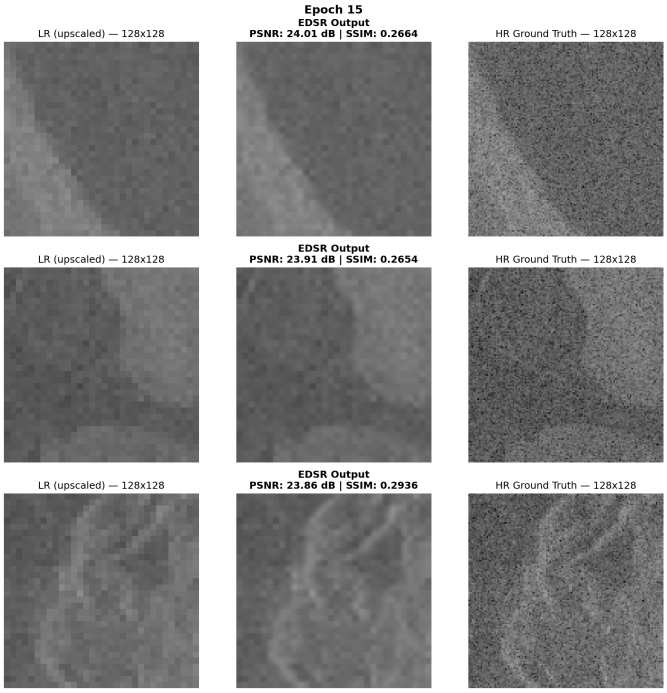

Fig. 4. SLC configuration at epoch 5. The HR ground truth contains significant speckle noise, and the EDSR output produces a smooth approximation that cannot match the noisy reference.

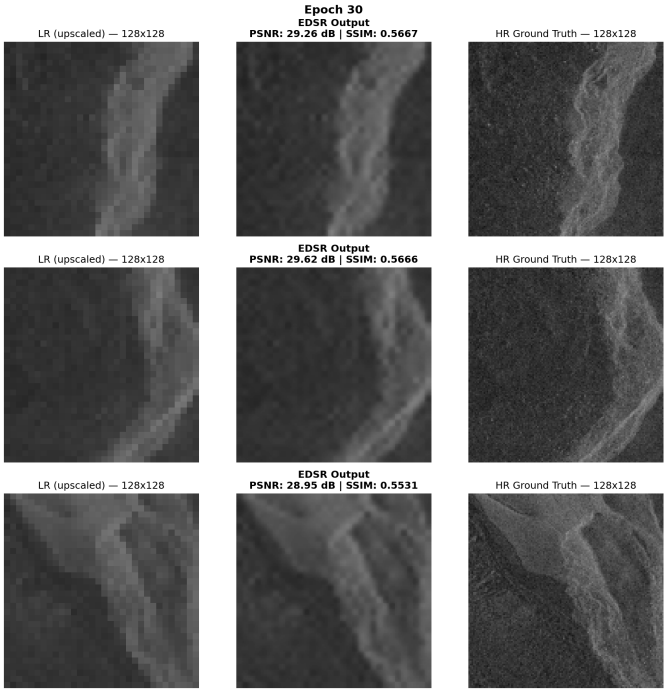

Fig. 5. Multilook configuration at epoch 500. The smoother ground truth enables the model to learn meaningful spatial structure. The EDSR output reconstructs terrain boundaries and geological features with visible improvement over the LR input.

In the SLC configuration the HR ground truth seems to appear visually noisier than both the LR input at the EDSR output illustrating the difficulty of using the raw data as a reference for training. The EDSR output is in this case performs implicit despeckling but cannot recover fine spatial details so details such as corners of objects or small details are not able to be captured.

In the multilook configuration the EDSR output shows clear improvement over the LR input especially in regions containing high contrast boundaries and the model successfully reconstructs sharper transitions between surface features that are blurred or features that are lost in the LR input. When looking at the patches that contain geological elements or sand dunes its apparent that corners and shapes start to appear more sharp or have more shape to them.

V. Discussion

When looking at the results it's apparent that multilook approach was critical to achieving a better result with contributing more than 5 dB PSNR improvement over the raw SLC training. This shows that the ground truth quality is equally important as model design in this self-supervised SAR super resolution.

PSNR and SSIM are imperfect metrics for this kind of image due to the speckled nature of SAR image introduces hard patterns and adds abstractions that are not really there, if an image that was normal but just blurry it would have an easier time to produce more accurate or sharper images. When looking at the images that the model create, metrics alone do not indicate and particular good reconstruction but on visual inspection the patches generated are quite good in terrain with high variance or high contrast.

For future work model architecture and more focus on multilook should be done in on multi sensor SAR images in order to overcome the dataset limitations that has been hit here.

VI. Conclusion

This project presented an EDSR based approach for 4x super resolution of Capella SAR SLC imagery using an multilook preprocessing approach. Key findings is that the multilook approach gave significant boost in metrics score and training, improving PSNR by 5.23 dB and SSIM by 0.3003. The EDSR model demonstrates capability of reconstructing SAR images from LR input and can replicate the terrain and high contrast features in the images. The primary limitation is the small single scene dataset that is homogeneous and a lot of preprocessing has to be done in order to get a good result. Future work should explore multi-scene training and SAR-specific evaluation metrics.

References

References

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